MATH310 Probability & Statistics
Section L2

Week 3: Lecture 06:
Discrete Random Variables

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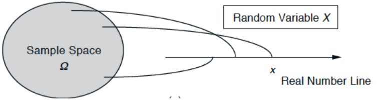
https://hulms.instructure.com/courses/3524

Random Variables

- So far we have talked about events and sample spaces.
- However for many experiments it is easier to use a summary variable.
- Say we are taking a opinion poll among 100 students about how understandable the lectures are.
- If "1" is used for understandable and "0" is used for not, then there are 2^{100} possible outcomes!!
- However, the thing that matters most is the number of students who think the class is understandable (or equivalently not).
- If we define a variable X to be that number, then the range of X is $\{0, 1, \ldots, 100\}$. **Much easier to handle that!**

Random Variable as a Mapping

Random Variable. A random variable is function from the sample space Ω into the real numbers.



Examples

- You toss a coin: is it head or tail?
- You roll a die: what number do you get?
- Number of heads in three coin tosses
- The sum of two rolls of die
- The number of die rolls it takes to get a six.

Discrete and continuous random variables

- A random variable is discrete if its range (the values it can take) is finite or at most countably finite.
  - $X =$ sum of two rolls of a die. $X \in \{2, \ldots, 12\}$.
  - $X =$ number of heads in 100 coin tosses. $X \in \{0, \ldots, 100\}$
  - $X =$ number of coin tosses to get a head. $X \in \{1, 2, 3, \ldots\}$
- Consider an experiment where you throw a dart which can fall anywhere between $[-1, 1]$.
- Let X be a the dart's position d. X is a random variable and X is not discrete.
- Now let Y be a random variable such that:

$$Y = \begin{cases} 1 & \text{If } d < 0 \\ 0 & \text{If } d \geq 0 \end{cases} \tag{2}$$

Now Y is a discrete random variable.

Probability Mass Functions

Earlier we talked about probability laws which assign probabilities to events in a sample space. One can play the same game with random variables.

- $X =$ number of heads in two fair coin tosses
- X can take values in $\{0, 1, 2\}$.
- $P(X = 0) =$
- $P(X = 1) =$
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- $P(X = 1) = P(\{HT, TH\}) = 1/2$.
- $P(X = 2) = P(\{HH\}) = 1/4$.
- In general, the probability that a random variable X takes up a value x is written as $p_X(x)$, or $P_X(x)$ or $P_X(X = x)$ etc.
- A random variable is always written with upper-case and the numerical value we are trying to evaluate the probability for is written with a lower case.

Properties of P.M.F's

- Perhaps not so surprisingly, $\sum_x P(X = x) = 1$
- $P(X \in S) = \sum_{x \in S} P(X = x)$
- To compute $P(X = x)$
  - Collect all possible outcomes that give $\{X = x\}$.
  - Add their probabilities to get $P(X = x)$.
- You are throwing 2 fair coins. What is the probability that you see at least one head?
  - Well at least one head is $\{HT, TH, HH\}$. So probability of this is $3/4$.

The Uniform random variable

Consider the roll of a fair die. You are interested in the number on the roll.

- X can take how many different values?
- Now what are the probabilities of taking on those values?
- $P(X = 1) = P(X = 2) = P(X = 3) = P(X = 4) = P(X = 5) = P(X = 6) = 1/6$
- For an **uniform** random variable X, each value has equal probability mass.
- If an **uniform** random variable X takes on k different values, then the probability mass at each of those values are $1/k$.

The Bernoulli random variable

Consider the toss of a biased coin, which gives a head with probability p.

- A **Bernoulli** random variable X takes two values: 1 if a head comes up and 0 if not. $X = \begin{cases} 1 & \text{If head} \\ 0 & \text{If tail} \end{cases}$
- The PMF is given by: $p_X(x) = \begin{cases} p & \text{If } x = 1 \\ 1 - p & \text{If } x = 0 \end{cases}$
- Examples of a Bernoulli
  - A person can be healthy or sick with a certain disease.
  - A test result for a disease can be positive or negative.
  - It may rain one day or not.

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What if the coin is biased? Probability of H is 0.8?

A biased coin

You are throwing 10 biased coins with $P(\{H\}) = p$. What is the probability that the sum X equals 5?

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- So $P(X = 5) =$
- What about $P(X = 8)$?

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- What about $P(X = 8)$? $\binom{10}{8} p^8(1-p)^2$.
- In general $P(X = k) = \binom{n}{k} p^k(1-p)^{n-k}$

The Binomial random variable

A biased coin ($P(\{H\}) = p$) is tossed n times. All tosses are mutually independent.

- Let X_i be a Bernoulli random variable which is 1 if the i^{th} toss gave a head. Then $\{X_i, i = 1, \ldots, n\}$ are independent Bernoullis.
- Let Y be the number of heads we see at the end of all n tosses.
- Whats the relationship of Y to the X_i's?

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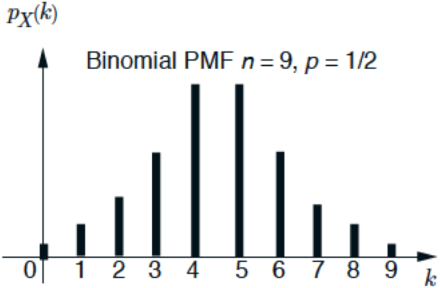
- $P(Y = n) = P(\{\text{all heads}\}) = p^n$.
- $P(Y = k) = \binom{n}{k} p^k (1 - p)^{n-k}$ for $k \in \{0, 2, \ldots n\}$
- $\sum_{k=0}^{n} \binom{n}{k} p^k (1 - p)^{n-k} = 1$ (Binomial expansion)!

The Binomial random variable

- Sum of independent Bernoullis give a Binomial!
- We will denote the Binomial PMF by *Binomial*(n, p) or *Bin*(n, p).
- We will write $X \sim \mathit{Binomial}(n, p)$ to indicate that X is **distributed** as a Binomial random variable with parameters n and p.
- Quick look at the histogram of $X \sim \mathit{Binomial}(9, 1/2)$.



Urn example

You have a urn with an 10 blue and 20 red balls. You pick 9 balls with replacement. Let X be the number of blue balls. What is $P(\text{X=3})$?

- With replacement is key here! I can draw the same ball twice!
- We have $P(\text{blue ball}) =$

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- X is a Binomial(9,3) random variable! $X \sim \mathit{Binomial}(9, 3)$

The Geometric random variable

- The Bernoulli PMF describes the probability of success/failure in a single trial.
- The Binomial PMF describes the probability of k successes out of n trials.
- Sometimes we may also be interested in doing trials until we see a success.

- Alice resolves to keep buying lottery tickets until he wins a hundred million dollars. She is interested in the random variable "number of lottery tickets bought until he wins the 100M$ lottery".
- Annie is trying to catch a taxi. How many occupied taxis will drive past before she finds one that is taking passengers?

- The number of trials required to get a single success is a **Geometric Random Variable**

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We repeatedly toss a biased coin $(P(\{H\}) = p)$. The geometric random variable is the number X of tosses to get a head.

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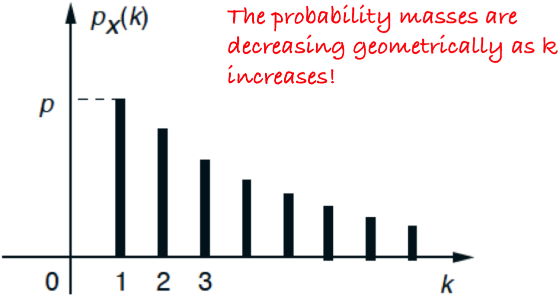
- $\sum_k P(X = k) = 1$ (why?)



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